

1.

Syntax :-

```
#include <iostream>
using namespace std;

int main ()
{
    // statements
    return 0 ;
}
```

2.

Conditions and loops :-↳ Conditional statements -

(i) if-else statement

```
if (condition)
{
    // statements
}
else (conditions)
{
    // statements
}
```

(iii) if statement

```
if (condition)
{
    // statements
}
```

(ii) Switch statement

```
switch (choice)
{
    case 1 :
        // statements
        break ;
    case 2 :
        // statements
        break ;
    default :
        cout << "Invalid choice"
}
```

↳ Loops -

(i) for loop

```
for (initialisation ; condition ; updation) {  
    // statements  
}
```

(ii) while loop

```
while (condition)  
{  
    // statements  
}
```

(iii) do-while loop

```
do  
{  
    // statements  
}  
while (condition);
```


3. Programming Problems

1. Square

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

Program:

```

#include <iostream>
using namespace std;

int main()
{
    for (int i=1 ; i<=5 ; i++)
    {
        for (int j=1 ; j<=5 ; j++)
        {
            cout << " * ";
        }
        cout << endl;
    }
    return 0;
}

```

2. Hollow Square

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

Program :

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
for (int i=1 ; i<=5 ; i++)
```

```
{
```

```
for (int j=1 ; j<=5 ; j++)
```

```
{
```

```
if (i==1 || i==5 || j==1 || j==5)
```

```
cout << " * " ;
```

```
else
```

```
cout << "   " ;
```

```
}
```

```
cout << endl ;
```

```
}
```

```
return 0 ;
```

```
}
```


3. Triangle

```

*
* *
* * *
* * * *
* * * * *

```

Program :

```

#include <iostream>
using namespace std;

int main ()
{
    for (int i=1; i<=5; i++)
    {
        for (int j=1; j<=i; j++)
        {
            cout << " * ";
        }
        cout << endl;
    }
    return 0;
}

```

4. Inverted Triangle

```
*****
****
***
**
*
```

Program :

```
#include <iostream>
using namespace std;
```

```
int main ()
{
```

```
    for (int i = 5 ; i >= 1 ; i--)
```

```
    {
        for (int j = 1 ; j <= i ; j++)
```

```
            cout << " * ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```


5.

Pyramid

```

      *
     **
    ***
   ****
  *****
 
```

Program:

#include <iostream>

using namespace std;

int main()

{

for (int i=1; i<=5; i++)

{

for (int j=1; j<=5-i; j++)

cout << " ";

for (int k=1; k<=2*i-1; k++)

cout << " *";

cout << endl;

}

return 0;

{

6.

Inverted Pyramid

```

* * * * *
 * * * * 
  * * * * 
   * * * * 
    * * * 
     * * 
      * 

```

Program :

```

#include <iostream>
using namespace std;

```

```

int main ()
{

```

```

    for (int i=5 ; i>=1 ; i--)
    {

```

```

        for (int j=1 ; j<=5-i ; j++)
            cout << " ";

```

```

        for (int k=1 ; k<=2*i-1 ; k++)
            cout << "*";

```

```

        cout << endl;

```

```

    }

```

```

    return 0;

```

```

}

```


Page 1

1. What is the purpose of the study?
2. What are the research questions?

2

$\epsilon = 10^{-1}$ (m) sind

11/10/2010

11/10/2010

$$f = \begin{pmatrix} i \\ j \\ k \end{pmatrix}, \quad \mu = \begin{pmatrix} i & j & k \end{pmatrix}$$
$$(-1, 1)$$

1. The first part of the text is a list of names and dates, which appears to be a record of some kind. The names are written in a cursive script, and the dates are in a more formal, printed style. The list includes names such as "John", "Mary", "James", and "Elizabeth", along with dates like "1810", "1811", and "1812".

1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840. 84

0.10/20.

8. Diamond

```

      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *

```

Program:

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
    int
    for (i=1 ; i<=5 ; i++)
    {
```

```
        for (int j=1 ; j<=5-i ; j++)
            cout << " ";
```

```
        for (int k=1 ; k<=2*i-1 ; k++)
            cout << "*";
```

```
        cout << endl;
```

```
    }
```

```
    for (int i=4 ; i>=1 ; i--)
```

```
    {
```

```
        for (int j=1 ; j<=5-i ; j++)
```

```
            cout << " ";
```

```
        for (int k=1 ; k<=2*i-1 ; k++)
```

```
            cout << "*";
```

```
    }
```

```
    cout << endl; }
```

```
    return 0;
```

```
}
```


code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=1;i<=5;i++)
7      {
8          for(int j=1;j<=5;j++)
9          {
10             cout<<"*";
11          }
12             cout<<endl;
13     }
14     return 0;
15 }
16
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\gaut2\Downloads\my code> g++ -g code code.cpp
PS C:\Users\gaut2\Downloads\my code> ./code.exe
```

```
PS C:\Users\gaut2\Downloads\my code>
```

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=1;i<=5;i++)
7      {
8          for(int j=1;j<=5;j++)
9          {
10             if(i==1||i==5||j==1||j==5)
11                 cout<<"*";
12             else
13                 cout<<" ";
14             }
15             cout<<endl;
16         }
17         return 0;
18     }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp
PS C:\Users\gaut2\Downloads\my code> ./code.exe
```

* *

* *

* *

```
PS C:\Users\gaut2\Downloads\my code>
```


code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=1;i<=5;i++)
7      {
8          for(int j=1;j<=i;j++)
9          {
10             cout<<"*";
11          }
12          cout<<endl;
13      }
14      return 0;
15  }
16
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp

PS C:\Users\gaut2\Downloads\my code> ./code.exe

*

**

PS C:\Users\gaut2\Downloads\my code> I

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=5;i>=1;i--)
7      {
8          for(int j=1;j<=i;j++)
9          {
10             cout<<"*";
11          }
12          cout<<endl;
13      }
14      return 0;
15  }
16
```

I

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp
PS C:\Users\gaut2\Downloads\my code> ./code.exe

**

*

code.cpp >

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=1;i<=5;i++)
7      {
8          for(int j=1;j<=5-i;j++)
9              cout<<" ";
10
11         for(int k=1;k<=2*i-1;k++)
12             cout<<"*";
13
14         cout<<endl;
15     }
16     return 0;
17 }
18
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp

PS C:\Users\gaut2\Downloads\my code> ./code.exe

```
*
***
*****
*****
*****
```

code.cpp > main()

```
4  int main()
5  {
6      for(int i=5;i>=1;i--)
7      {
8          for(int j=1;j<=5-i;j++)
9              cout<<" ";
10
11         for(int k=1;k<=2*i-1;k++)
12             cout<<"*";
13
14         cout<<endl;
15     }
16     return 0;
17 }
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp
PS C:\Users\gaut2\Downloads\my code> ./code.exe

*

code.cpp > ...

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      for(int i=1;i<=5;i++)
7      {
8          for(int j=1;j<=i;j++)
9              cout<<"*";
10
11         cout<<endl;
12     }
13     for(int i=4;i>=1;i--)
14     {
15         for(int j=1;j<=i;j++)
16             cout<<"*";
17
18         cout<<endl;
19     }
20     return 0;
21 }
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\gaut2\Downloads\my code> ./code.exe

```
*
**
***
****
*****
****
***
**
*
```

code.cpp > main()

```
4  int main()
6      for(int i=1;i<=5;i++)
16     for(int i=4;i>=1;i--)
17     {
18         for(int j=1;j<=5-i;j++)
19             cout<<" ";
20
21         for(int k=1;k<=2*i-1;k++)
22             cout<<"*";
23
24         cout<<endl;
25     }
26
27
28     return 0;
29 }
30
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\gaut2\Downloads\my code> g++ -o code code.cpp
PS C:\Users\gaut2\Downloads\my code> ./code.exe

```
*
***
*****
*****
*****
*****
*****
***
*
```

I